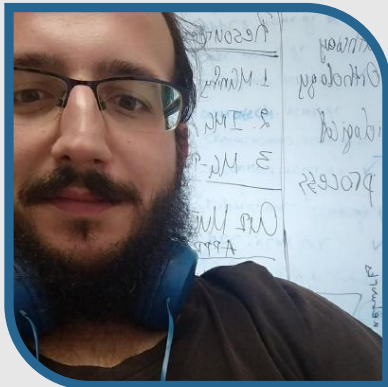




Zafeiropoulos Haris

Bioinformatics, Systems Biology, Microbial Ecology

-  April 24, 1991
-  Groefstraat 17, Leuven, Belgium
-  +30 694 909 3089
-  <https://hariszaf.github.io/>
-  haris.zafr@gmail.com

About Me

I develop bioinformatics approaches to better understand microbial communities, their dynamics and microbial interactions. My main scientific focus is on microbial interactions—especially metabolic ones—and their evolutionary implications. I am particularly interested in how microbial communities are shaped by their environment and, in turn, how their structure feeds back to influence that environment. To address these questions, I use diverse approaches, spanning 'omics, network analysis, and metabolic modeling.

Social Network

-  [Github Profile](#)
-  [LinkedIn Profile](#)
-  [Google Scholar Profile](#)
-  [ResearchGate Profile](#)

Languages

-  Greek ● ● ● ● ●
-  English ● ● ● ● ●
-  French ● ● ● ● ●

Research projects - working experience

Formal Employment Projects

- 2022 - 2026 **3D'omics** Post-doc
Aim of the project is to come up with technology and protocols to generate 3D omics landscapes, achieving reconstructions of intestinal host microbiota ecosystems. My role on the project is the analysis of microbial networks integrating spatial and metabolic data to predict species interactions and/or influence of metabolites on species distributions using co-occurrence network analysis and Genome-Scale Metabolic Modeling.
- 2022 - 2022 **A workflow for marine Genomic Observatories data analysis** Research associate
An EOSC-Life-supported project aimed at making the large volumes of data produced by genomic observatories more easily interpretable by providing the taxonomic inventories of each sample in a timely and non-technical format. I was the lead developer of the delivered workflow, and the Principal Investigator in the second half of the project.
- 2018 - 2021 **PREGO: Process, environment, organism (PREGO)** PhD
PREGO is a systems-biology approach to elucidate ecosystem function at the microbial dimension. PREGO was the main project of my PhD and my role was to develop the data integration parts of the platform.
- 2019 – present **ELIXIR-GR** technical support
ELIXIR-GR is the Greek National Node of the ESFRI [European RI ELIXIR](#), a distributed e-Infrastructure aiming at the construction of a sustainable European infrastructure for biological information. My role on the project was to develop containerized-versions of various software tools.
- 2018 - 2020 **RECONNECT** PhD
RECONNECT is an Interreg V-B "Balkan-Mediterranean 2014-2020" project. It aims to develop strategies for sustainable management of Marine Protected Areas (MPAs) and Natura 2000 sites.

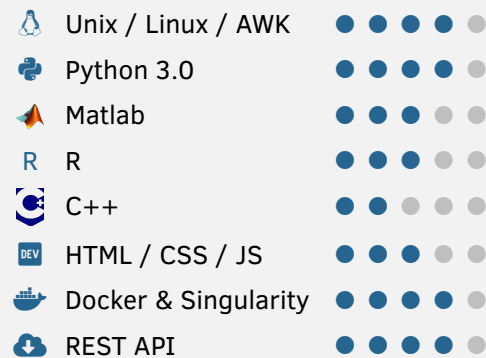
Unofficial Contributions

- 2021 - 2023 **EMBRC** PhD
The European Marine Biological Resource Centre (EMBRC) was established in 2013 to advance fundamental and applied marine biology and ecology research, while promoting sustainable blue economy. Over the last years, I have contributed in tasks of the [Greek node](#) in the development of workflows for the analysis of the data produced by the network.
- 2020 - ongoing **LifeWatch-ERIC** PhD
LifeWatch ERIC is a European Research Infrastructure Consortium providing e-Science research facilities to scientists investigating biodiversity and ecosystem functions and services in order to support society in addressing key planetary challenges. Since PEMA was selected to be part of an [Internal Joint Initiative](#), I have been maintaining the workflow to address the needs of the infrastructure.

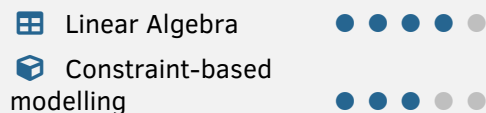
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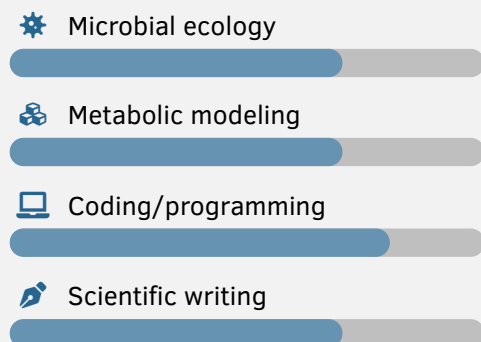
Coding Skills



Math and Stats



Barskills



Education

Graduate studies

2018 – 2022	PhD in Bioinformatics	University of Crete, Biology department
	Dissertation	IMBBC - HCMR
	Microbial communities through the lens of high throughput sequencing, data integration and metabolic networks analysis	
2016 – 2018	M.Sc. in Bioinformatics	University of Crete, School of Medicine
	grade: 9.1/10.0	
	Master Thesis	IMBBC - HCMR
	eDNA metabarcoding for biodiversity assessment: Algorithm design and bioinformatics analysis pipeline implementation	

Undergraduate studies

2011 – 2016	B.Sc. in Biology	National & Kapodistrian University of Athens
	grade: 6.2 / 10.0	
	Bachelor Thesis	School of Science, department of Biology
	Morphology, morphometry and anatomy of species of the genus <i>Pseudamnicola</i> in Greece	

Publications

2026	Sampling from the Solution Space and Metabolic Environments of Genome-Scale Metabolic Models <i>Zafeiropoulos, Haris and Rios Garza, Daniel</i> arXiv, DOI: 10.48550/arXiv.2603.29546	
2025	microbetag: simplifying microbial network interpretation through annotation, enrichment tests, and metabolic complementarity analysis <i>Zafeiropoulos, Haris, Delopoulos, Ermis Ioannis Michail, Erega, Andi, Schneider, Aline, Geirnaert, Annelies, Morris, John and Faust, Karoline</i> Genome Biology, DOI: 10.1186/s13059-025-03769-2	
2025	Establishing the ELIXIR Microbiome Community <i>Finn, R. D., Balech, B., Burgin, J., Chua, P., Corre, E., Cox, C. J., Donati, C., dos Santos, V. M., Fosso, B., Hancock, J., Heil, K. F., Ishaque, N., Kale, V., Kunath, B. J., Médigue, C., Nogueira, T., Pafilis, Evangelos, Pesole, G., Richardson, L., Santamaria, M., Strepis, N., Van Den Bossche, T., Vizcaíno, J. A., Zafeiropoulos, Haris, Willassen, N. P., Pelletier, E. and Batut, B.</i> F1000Research, DOI: 10.12688/f1000research.144515.2	
2025	μGrowthDB: querying, visualizing, and sharing microbial growth curve data <i>Radev, Andrey, Casado Gómez-Pallete, Julia, Monsalve Duarte, Sofia, Faust, Karoline and Zafeiropoulos, Haris</i> bioRxiv, DOI: 10.1101/2025.10.13.682118	
2025	A long-term ecological research data set from the Marine Genetic Monitoring Program ARMS-MBON 2018–2020 <i>Daraghmeh, Nauras, Exter, Katrina, Pagnier, Justine, Balazy, Piotr, Cancio, Ibon, Chatzigeorgiou, Giorgos, Chatzinikolaou, Eva, Chelchowski, Maciej, Christmas, Nathan Alexis Mitchell, Comtet, Thierry and others</i> Molecular Ecology Resources, DOI: 10.1111/1755-0998.14073	
2024	Predicting microbial interactions with approaches based on flux balance analysis: an evaluation <i>Joseph, Clémence, Zafeiropoulos, Haris, Bernaerts, Kristel and Faust, Karoline</i> BMC Bioinformatics, DOI: 10.1186/s12859-024-05651-7	

- 2024 **Improving genome-scale metabolic models of incomplete genomes with deep learning**
Boer, Meine D., Melkonian, Chrats, **Zafeiropoulos, Haris**, Haas, Andreas F., Rios Garza, Daniel and Dutilh, Bas E.
iScience, DOI: [10.1016/j.isci.2024.111349](https://doi.org/10.1016/j.isci.2024.111349)
- 2024 **dingo: a Python package for metabolic flux sampling**
Chalkis, Apostolos, Fisikopoulos, Vissarion, Tsigaridas, Elias and **Zafeiropoulos, Haris**
Bioinformatics Advances, DOI: [10.1093/bioadv/vbae037](https://doi.org/10.1093/bioadv/vbae037)
- 2024 **Unveiling emerging opportunistic fish pathogens in aquaculture: a comprehensive seasonal study of microbial composition in Mediterranean fish hatcheries**
Sklaros, Dimitrios, Kostakou, Maria, Kokkari, Constantina, Tsertou, Maria Ioanna, Pavloudi, Christina, **Zafeiropoulos, Haris**, Katharios, Pantelis and Flietakis, Emmanouil
Microorganisms, DOI: [10.3390/microorganisms12112281](https://doi.org/10.3390/microorganisms12112281)
- 2023 **metaGflow: a workflow for the analysis of marine Genomic Observatories shotgun metagenomics data**
Zafeiropoulos, Haris, Beracochea, Martin, Ninidakis, Stelios, Exter, Katrina, Potirakis, Antonis, De Moro, Gianluca, Richardson, Lorna, Corre, Erwan, Machado, João, Pafilis, Evangelos and others
GigaScience, DOI: [10.1093/gigascience/giad078](https://doi.org/10.1093/gigascience/giad078)
- 2023 **Metabolic models of human gut microbiota: advances and challenges**
Rios Garza, Daniel, Gonze, Didier, **Zafeiropoulos, Haris**, Liu, Bin and Faust, Karoline
Cell Systems, DOI: [10.1016/j.cels.2022.11.002](https://doi.org/10.1016/j.cels.2022.11.002)
- 2022 **PREGO: a literature and data-mining resource to associate microorganisms, biological processes, and environment types**
Zafeiropoulos, Haris, Paragkamian, Savvas, Ninidakis, Stelios, Pavlopoulos, Georgios A., Jensen, Lars Juhl and Pafilis, Evangelos
Microorganisms, DOI: [10.3390/microorganisms10020293](https://doi.org/10.3390/microorganisms10020293)
- 2022 **Deciphering the community structure and the functional potential of a hypersaline marsh microbial mat community**
Pavloudi, Christina and **Zafeiropoulos, Haris**
FEMS Microbiology Ecology, DOI: [10.1093/femsec/fiac141](https://doi.org/10.1093/femsec/fiac141)
- 2022 **Automating the curation process of historical literature on marine biodiversity using text mining: the DECO workflow**
Paragkamian, Savvas, Sarafidou, Georgia, Mavraki, Dimitra, Pavloudi, Christina, Beja, Joana, Eliezer, Menashé, Lipizer, Marina, Boicenco, Laura, Vandepitte, Leen, Perez-Perez, Ruben and others
Frontiers in Marine Science, DOI: [10.3389/fmars.2022.940844](https://doi.org/10.3389/fmars.2022.940844)
- 2021 **0s and 1s in marine molecular research: a regional HPC perspective**
Zafeiropoulos, Haris, Gioti, Anastasia, Ninidakis, Stelios, Potirakis, Antonis, Paragkamian, Savvas, Angelova, Nelina, Antoniou, Aglaia, Danis, Theodoros, Kaitetzidou, Eliza, Kasapidis, Panagiotis and others
GigaScience, DOI: [10.1093/gigascience/giab053](https://doi.org/10.1093/gigascience/giab053)
- 2021 **The Dark mAtter iNvestigator (DARN) tool: getting to know the known unknowns in COI amplicon data**
Zafeiropoulos, Haris, Gargan, Laura, Hintikka, Sanni, Pavloudi, Christina and Carlsson, Jens
Metabarcoding and Metagenomics, DOI: [10.3897/mbmg.5.69657](https://doi.org/10.3897/mbmg.5.69657)
- 2021 **Geometric Algorithms for Sampling the Flux Space of Metabolic Networks**
Chalkis, Apostolos, Fisikopoulos, Vissarion, Tsigaridas, Elias and **Zafeiropoulos, Haris**
37th International Symposium on Computational Geometry (SoCG 2021), DOI: [10.4230/LIPIcs.SocG.2021.21](https://doi.org/10.4230/LIPIcs.SocG.2021.21)
- 2021 **The Santorini Volcanic Complex as a valuable source of enzymes for bioenergy**
Polymenakou, Paraskevi N., Nomikou, Paraskevi, **Zafeiropoulos, Haris**, Mandalakis, Manolis, Anastasiou, Thekla I., Kiliadis, Stephanos, Kyrpides, Nikos C., Kotoulas, Georgios and Magoulas, Antonios
Energies, DOI: [10.3390/en14051414](https://doi.org/10.3390/en14051414)

2020 **PEMA: a flexible pipeline for environmental DNA metabarcoding analysis of the 16S/18S ribosomal RNA, ITS, and COI marker genes**
Zafeiropoulos, Haris, Viet, Ha Quoc, Vasileiadou, Katerina, Potirakis, Antonis, Arvanitidis, Christos, Topalis, Pantelis, Pavloudi, Christina and Pafilis, Evangelos
 GigaScience, DOI: [10.1093/gigascience/giaa022](https://doi.org/10.1093/gigascience/giaa022)

Conferences

2025	Poster presentation: \texttt{microbetag}: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis 24th Annual International Conference on Systems Biology (ICSB 2025) @ Dublin, Ireland	ICSB25
2024	Poster presentation: \texttt{microbetag}: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis 19th International Symposium on Microbial Ecology @ Cape Town, South Africa	ISME19
2024	Poster presentation: \texttt{microbetag}: simplifying microbial network interpretation through annotation, enrichment and metabolic complementarity analysis Belgian Society for Microbiology: Annual Symposium @ Brussels, Belgium	BSM2024
2022	Poster presentation: \texttt{microbetag}: a microbial co-occurrence network annotator 1st Applied HoloGenomics conference @ Bilbao, Spain	AHC2022
2022	Poster presentation: Reverse ecology and systems biology approaches to identify key processes ensuring life in extreme environments 18th International Symposium on Microbial Ecology @ Lausanne, Switzerland	ISME18
2021	Flash talk: \texttt{dingo}: A python library for metabolic networks sampling & analysis Bioinformatics Open Source Conference (BOSC) - online	BOSC2021
2021	Flash talk: PEMA v2: addressing metabarcoding bioinformatics analysis challenges 1st DNAQUA International Conference - online	DNAQUA2021
2020	Flash talk: Mining literature and -omics (meta)data to associate microorganisms, biological processes and environment types Federation of European Microbiological Societies (FEMS) - online	FEMS2020
2020	Oral presentation: Geometric and statistical methods in systems biology: the case of metabolic networks PyData Global - online	PyData2020
2019	Participation network2019: 4th Symposium on Ecological Networks @ Paris, France	Network2019
2019	ePoster presentation: P.E.M.A.: a Pipeline for Environmental DNA Metabarcoding Analysis 8th International Barcode of Life Conference @ Trondheim, Norway	iBOL2019
2018	Poster presentation: P.E.M.A.: a Pipeline for Environmental DNA Metabarcoding Analysis European Conference on Computational Biology (ECCB) 2018 @ Athens, Greece	ECCB18

Workshops & Summer schools

2025	School of artificial intelligence applied to microbiomes @ AgroParisTech, France - Sampling the solution space of metabolic networks (theory & hands-on)	SAIM
2025	Data integration with Microbial Networks and Community Models @ KULeuven, Belgium - <i>Organizing committee</i> - microbetag: metabolic secrets behind microbial co-occurrence (theory & hands-on)	DIMNCM
2025	Metabolic models applied to microbiomes @ Antony, France - <i>Organizing committee</i> - Metabolic model analysis methods (theory & hands-on)	MMAM
2023	Summer School on Microbial Community Design and Control @ Leuven, Belgium Assisting on "metabolic network reconstruction" topic	DCMC
2023	Economic Principles in Cell Physiology @ Paris, France Mathematical modeling of cellular systems and "resource allocation thinking". <i>Poster presentation</i> : "saml'em all": exploring potential flux sampling applications	EPCP2023
2022	Microbial communities: current approaches and open challenges @ Cambridge, UK Interdisciplinary exchanges in microbial communities' research and the emerging challenge areas.	UMCW06
2022	ELIXIR Fluxomics Training School 2021 An introduction to the field of fluxomics and the experimental and computational methods used to estimate and predict metabolic fluxes	ELIXFLUX
2020	Metagenomics, Metatranscript- omics and multi 'omics for microbial community studies A Physalia Course from Prof. Dr. Curtis Huttenhower	material

Supervision

2025–ongoing	Sourav Pattanaik, MSc student "Populating μ GgrowrhDB"	
2024–ongoing	Sotiris Touloupoulos, ERASMUS student "Pre- and post-sampling features to leverage flux sampling at both the strain and the community level"	GitHub repo
2024–ongoing	Xi Peng, Visiting PhD candidate "Enhancing microbetag's functional annotations with quorum sensing prediction capabilities"	
2024–ongoing	Anna Voukouna, MSc student "Investigation of microbial metabolic interactions potential across the Genome Taxonomy Database representative genomes"	
2024–2025	Andrey Radev, MSc student "Visualisation and analysis of microbial growth data to facilitate interactions investigation and interactions' strength assessment"	
2024	Sotiris Touloupoulos, GSoC student "Pre- and post-sampling features to leverage flux sampling at both the strain and the community level"	blogpost
2023–2024	Sofia Monsalve Duarte, MSc student "Creation and implementation of a database interface for the collection and analysis of microbial growth data"	

2023–2024	Ermis Ioannis Michail Delopoulos, GSoC student post "Development of a Cytoscape App for microbe-microbe association networks"
2022–2024	Ermis Ioannis Michail Delopoulos, MSc student "Development of a graphical user interface for microbetag"
2022–2023	Julia Casado Gómez-Pallete, MSc student thesis "Development of a database for human gut bacterial growth curves"

Teaching

2025	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2025	Introduction in metabolic modeling GitHub repo Theoretical and practical session on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2024	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective (part II) slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2024	Exploring the microbiome with high-throughput sequencing technologies: a bioinformatics perspective slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2023	eDNA metabarcoding, pipeline development & high performance computing slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2023	eDNA metabarcoding, pipeline development & high performance computing slides Lecture on MSc "Applied Bioinformatics & Data Analysis" in Democritus University of Thrace , Greece
2022	Genome-scale model reconstruction GColab notebook Lectures on "Master en bioinformatique et modélisation" of the Université Libre de Bruxelles
2021	Microbiome Data Analyses Workshop Online MDAWO Tutorial: PEMA: a flexible Pipeline for Environmental DNA Metabarcoding Analysis of the 16S/18S rRNA, ITS and COI marker genes.
2020	Environmental DNA & DNA metabarcoding slides Lecture on MSc "Environmental Biology" in University of Crete
2019	8th Conference of Hellenic Society of Mikrobiokosmos MKB8 Tutorial: PEMA: a flexible Pipeline for Environmental DNA Metabarcoding Analysis of the 16S/18S rRNA, ITS and COI marker genes.

Awards & Scholarships

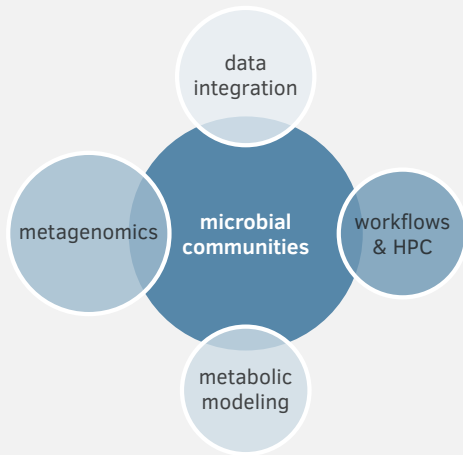
Zafeiropoulos Haris

Bioinformatics, Systems
Biology, Microbial Ecology

Software

pema prego darn dingo
microbetag

Research fields



Memberships



- 2021 **9th conference of the Hellenic Society of Mikrobiokosmos** [MBK9](#)
Oral presentation award: Sampling the flux space of microbial metabolic networks for (un)biased assessment of the possible functional states: the example of Sars-Cov-2 on the human alveolar macrophage metabolic network.
- 2021 **Google Summer of Code** [GSoC](#)
Project title: *From DNA sequences to metabolic interactions: building a pipeline to extract key metabolic processes*
- 2021 **European Molecular Biology Organization Short-Term Fellowship** [EMBO](#)
Project title: *"Exploiting data integration, text-mining and computational geometry to enhance microbial interactions inference from co-occurrence networks"*
- 2020 **Federation of European Microbiological Societies Meeting Attendance Grant** [FEMS](#)
for joining the *"Metagenomics, Metatranscript- omics and multi 'omics for microbial community studies"* Physalia course
- 2019 **Short Term Scientific Mission (STSM) @ DNAqua-net COST action** [DNAqua-net](#)
Project title *A comparison of bioinformatic pipelines and sampling techniques to enable benchmarking of DNA metabarcoding.* [Report.](#)
- 2018 **Best Poster Award @ Hellenic Bioinformatics conference** [HBIO](#)
for *PEMA: a Pipeline for Environmental DNA Metabarcoding Analysis*

References

- Karoline Faust** Department of Microbiology, Immunology and Transplantation, Rega Institute for Medical Research, Laboratory of Molecular Bacteriology, KU Leuven Leuven, 3000, Leuven, Belgium
📞: +3216322698 | @: karoline.faust@kuleuven.be
- Pafilis Evangelos** Institute of Marine Biology, Biotechnology & Aquaculture, Hellenic Centre for Marine Research, (IMBBC - HCMR), Gournes, Pediados, P.O. Box 2214, Heraklion Crete 71003, Greece
📞: +30 2810 337740 | @: pafilis@hcmr.gr
- Georgios Kotoulas** Institute of Marine Biology, Biotechnology & Aquaculture, Hellenic Centre for Marine Research, (IMBBC - HCMR) Gournes, Pediados, P.O. Box 2214, Heraklion Crete 71003, Greece
📞: +30 2810 337740 | @: kotoulas@hcmr.gr